MACHINE LEARNING vs. DEEP LEARNING

REAL-WORLD APPLICATIONS

EXAMPLE 1: MACHINE LEARNING

HOUSE PRICE PREDICTION

Real-world application:

Zillow

Zestimate

VS.

WHY MACHINE LEARNING IS SUITABLE

Zillow's Zestimate uses machine learning models to estimate home values based on structured data such as:

- Number of bedrooms and bathrooms
- Square footage
- Property age
- Neighborhood
- Recent home sales
- Property taxes and other public records

House price prediction is a regression problem where the important features are already known. Machine learning algorithms, such as Linear Regression and Gradient Boosting, can efficiently learn the relationship between these variables and housing prices. These models are relatively easy to train, require less computational power, and produce accurate predictions for structured datasets.

WHY DEEP LEARNING IS LESS SUITABLE

Deep learning generally performs best with large amounts of unstructured data, such as images, audio, or text. For structured housing data, deep learning often adds unnecessary complexity while providing only modest improvements in prediction accuracy. It also requires more computing resources and is less interpretable than traditional machine learning models.

EXAMPLE 2: DEEP LEARNING

NATURAL LANGUAGE PROCESSING (MACHINE TRANSLATION)

Real-world application:

Google Translate

Hello, how are you? → Bonjour, comment allez-vous?

WHY DEEP LEARNING IS SUITABLE

Google Translate uses deep learning, specifically Transformer neural networks, to translate text between hundreds of languages. Unlike traditional machine learning, deep learning models automatically learn grammar, sentence structure, word relationships, and context directly from massive amounts of multilingual text.

English Sentence → Transformer Model → Translated Sentence

For example, when translating an English sentence into French, the model considers the meaning of the entire sentence rather than translating each word individually. This allows it to produce more natural and accurate translations.

Deep learning is well suited for this task because language is highly complex and context dependent. Transformer models can process long sequences of text and understand relationships between words that traditional machine learning algorithms struggle to capture.

WHY TRADITIONAL MACHINE LEARNING IS LESS SUITABLE

Traditional machine learning requires manually engineered linguistic features, such as grammar rules and dictionaries, which are difficult to develop and maintain across many languages. It also struggles to understand context, idioms, and sentence meaning, resulting in less accurate translations than deep learning models.

CONCLUSION

Machine learning and deep learning each have strengths depending on the problem being solved.

MACHINE LEARNING

Ideal for structured datasets where relevant features are already known.

Example: House Price Prediction

Uses known variables like location, square footage, and number of bedrooms to accurately estimate property values with relatively simple models.

KEY DIFFERENCES

Data Type	Structured (Tabular data)	Unstructured (Text, Images, Audio)
Feature Extraction	Manual (Engineered by humans)	Automatic (Learned by model)
Data Requirement	Smaller datasets	Large datasets
Computational Resources	Lower	Higher
Interpretability	More interpretable	Less interpretable

DEEP LEARNING

Better suited for complex, unstructured data that requires automatic feature extraction.

Example: Google Translate

Transformer models learn language patterns and context directly from large datasets, enabling highly accurate translations across many languages.

Choosing the appropriate approach depends on the type of data, the complexity of the task, and the available computational resources.

Machine Learning vs. Deep Learning: Real-World Applications

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Example 1: Machine Learning – House Price Prediction

Real-world application: Zillow's home value estimation system (Zestimate).

Why machine learning is suitable:

House price prediction is a regression problem where structured historical data is available. Machine learning algorithms such as Linear Regression analyze features including:

- Number of bedrooms and bathrooms
- Square footage
- Property age
- Neighborhood
- Lot size
- Nearby schools and amenities

These features are manually selected because they have a direct impact on property values. The model learns the relationship between these variables and house prices from past sales, allowing it to estimate the value of new properties. Machine learning is efficient, interpretable, and performs well with structured tabular data.

Why deep learning is less suitable:

Although deep learning can predict house prices, it typically requires much larger datasets and significantly more computing power. For structured data like housing records, deep learning often provides only small improvements in accuracy while making the model more difficult to train and interpret. Since the important features are already known, traditional machine learning is usually the more practical and cost-effective choice.

Example 2: Deep Learning – Natural Language Processing (Machine Translation)

Real-world application: Google Translate

Why deep learning is suitable:

Google Translate uses deep learning through Transformer-based neural machine translation models, building on the earlier Google Neural Machine Translation (GNMT) system, to translate text between hundreds of languages. Unlike traditional machine learning, deep learning models automatically learn grammar, sentence structure, word relationships, and context directly from massive amounts of multilingual text. For example, when translating an English sentence into French, the model considers the meaning of the entire sentence rather than translating each word individually. This allows it to produce more natural and accurate translations. Deep learning is well suited for this task because language is highly complex and context dependent. Transformer models can process long sequences of text and understand relationships between words that traditional machine learning algorithms struggle to capture.

Why traditional machine learning is less suitable:

Traditional machine learning requires manually engineered linguistic features, such as grammar rules and dictionaries, which are difficult to develop and maintain across many languages. It also struggles to understand context, idioms, and sentence meaning, resulting in less accurate translations than deep learning models.

Machine learning and deep learning are both powerful branches of artificial intelligence, but they are best suited for different types of problems.

- Machine learning is ideal for structured datasets where important features are already known. House price prediction is a good example because models such as Linear

Regression can accurately estimate home values using variables like square footage, location, and number of bedrooms while requiring relatively little computational power.

- Deep learning is better suited for complex, unstructured data that requires automatic feature extraction. Google Translate is an excellent example because Transformer models learn language patterns and context directly from large datasets, enabling highly accurate translations across many languages.

The choice between machine learning and deep learning depends on the type of data, the complexity of the task, and the computational resources available.

AI Use Statement: I used ChatGPT-5.5 to improve the clarity and organization of my writing, refine my explanations, and perform final editing after completing my initial draft. I also used Scribbr's APA Citation Generator to generate and verify references in accordance with APA 7th edition guidelines. All ideas and conclusions are my own, and I reviewed and verified all AI-assisted content for accuracy before submission.

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